

Learning When Not to Inspect: Risk-Calibrated Weak Supervision for Efficient Industrial Visual Inspection

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Abstract

Inline visual inspection is dominated by normal images: most of the compute a production line spends goes to confirming that nothing is wrong. We ask whether a lightweight scout trained only with image-level OK/NOK labels — no boxes, no masks — can learn when *not* to inspect: safely discarding a large fraction of normal images before an expensive detector runs, under an explicit budget on missed defects. We present a three-component selective-inspection cascade — a weakly supervised scout, a risk-calibrated early-exit stage, and a strong pluggable inspector — completed by a subordinated conformal accept/review/reject decision layer whose strictness is adjustable after deployment without retraining. The early exit, calibrated under an operator-chosen missed-defect budget and then held fixed, removes up to 42.2% of normal images from the expensive path while keeping a measured exit-stage defect recall of 0.9947, cutting mean per-image inspection cost accordingly — a throughput gain, with tail latency unchanged, and we report it as such. A supervision-budget study answers the practical question of how many defect labels are enough: the scout is near chance below fifty labelled defects, becomes reliably usable above two hundred, and is seed-dependent in between. The cascade, its calibration rule, and its measured integration costs are reported with every negative finding disclosed.

1. Introduction

1.1 The question

A deployed inspection line sees an overwhelmingly normal image stream: defective parts are rare, and the strongest available defect detectors are paid in full on every image, defective or not. At the same time the labels a line actually accumulates are lopsided — OK images arrive essentially for free, image-level NOK labels slowly, and localized annotations (boxes, masks) only at deliberate labelling cost (Božič, Tabernik, and Skočaj 2021). Recent critical analyses of industrial anomaly detection argue that leaderboard accuracy under academic protocols answers few of the questions a production deployment asks (Baitieva et al. 2025); the question we study here is one of the unanswered ones, and it is falsifiable: *can a scout trained only with image-level OK/NOK labels safely discard a large fraction of normal images before running an expensive inspector, under a calibrated missed-defect budget?*

Our thesis is that in OK-dominated industrial flows, a few image-level NOK labels suffice to learn a calibrated early-exit policy that cuts the mean inspection cost without materially degrading defect recall. The mechanism is a three-component cascade (Figure 1): a **weak scout** trained from image labels alone scores every incoming image; a **risk-calibrated Fast-OK-Exit** admits confidently normal images immediately, with its exit threshold calibrated on validation under an operator-chosen missed-NOK budget and applied frozen; the remainder is forwarded to a **strong inspector**. The inspector slot is deliberately external and pluggable: we claim no improvement over any detector’s raw accuracy, and the strongest plug-in we measured is a fully supervised external detector (Robinson et al. 2026) at 0.9977 image-AUROC. The scout-side metrics — exit rate and exit-stage defect recall — are independent of which inspector is plugged in; the integrated end-to-end numbers we report use our own residual-DETR head as the inspector instantiation, and we state that choice, and its cost, explicitly where those numbers appear (§6). A subordinated conformal accept/review/reject decision layer completes the cascade, converting any scorer’s output into a three-way decision with a finite-sample false-reject guarantee (§7).

1.2 Contributions

C1 — Risk-calibrated, weakly supervised early exit. An exit policy trained from image-level labels only and calibrated under an explicit missed-NOK budget on validation, applied frozen to test: 38.1%

(zero-budget) and 42.2% ($\leq 1\%$ budget) of OK images exit before the inspector at a measured exit-stage NOK-recall of 0.9947 (Clopper–Pearson 95% CI [0.9707, 0.9999]) (§6.1).

C2 — The supervision-budget curve. A controlled answer to “how many defect labels are enough”: with the OK pool fixed, image-AUROC moves from near chance at ≤ 50 NOK labels through a seed-dependent transition near 100 to reliable operation at ≥ 200 , replicated across three training seeds with the transition point’s seed dependence disclosed (§5.2).

C3 — The measured end-to-end risk–compute trade-off. The integrated cascade, measured as one system under a single strict-serial latency protocol: mean per-image cost falls by 15.1–16.7% at the two exit budgets while p95 latency is unchanged — a throughput gain, not a detection-latency gain, reported as such — and the integration costs exactly one exit-stage-missed defect relative to running the inspector on everything (§6.2–§6.3).

C4 — A certifiable conformal decision layer, subordinated to the cascade. A split-conformal accept/review/reject rule over image scores with a finite-sample marginal false-reject guarantee, certifiable at $\alpha_{\text{reject}} = 0.005$ on our calibration set and applied identically over four scorers, own and external — acceptance strictness becomes a post-hoc deployment knob (§7).

Every claim above is bound to the dataset, surface, and protocol of §4, and the paper’s negative findings — the scout’s near-chance floor below fifty NOK labels, the seed-dependent knee, the one-defect integration cost, the unchanged tail latency, and the regime where a label-free baseline is not cleared — are disclosed alongside the positives they qualify.

2. Problem setting

Task and interface. Given an image I , an inspection system must decide whether the part is normal (OK) or anomalous (NOK) under a real-time budget. Every model in this paper — scout, inspector, and every baseline — reduces to a common interface: a per-image anomaly score $\hat{a}(I) \in \mathbb{R}$, optionally with localisation output. Threshold-free ranking metrics are computed directly on $\hat{a}$; every thresholded decision derives from the same stored per-image scores.

Label regimes. The scout’s supervision is the regime a production line actually reaches first: image-level OK/NOK labels, with no boxes and no masks. We evaluate it in two label pools: a **few-NOK** pool (3 088 images: 2 361 OK / 727 NOK), reflecting the scarce-defect stream a line accumulates, and a **balanced** pool (1 456 images: 729 OK / 727 NOK), reflecting a curated retraining set. Two reference regimes bound the axis: **label-free** OK-only anomaly detection (Roth et al. 2022), which consumes no NOK labels at all and therefore sets the floor any weak-label method must justify itself against, and **full supervision** with boxes or masks on all defects, the regime the pluggable inspector consumes.

The selective-inspection objective. Let c_s and c_i be the per-image costs of the scout and the inspector, and let $e \in [0, 1]$ be the fraction of images that exit early. The cascade’s expected per-image cost is $c_s + (1 - e)c_i$: the scout is paid on every image, so the exit buys mean cost only when c_s is small against c_i and e is large. The safety constraint is a **missed-NOK budget** B : the exit threshold must be chosen so that the fraction of defective images wrongly exited, estimated on validation, does not exceed B . This is a risk–compute trade-off, not a pure speedup: we report both sides of it, including the tail-latency side the mean conceals (§6.3).

Metrics. Image-level AUROC (NOK positive) is the headline ranking metric. The exit stage reports OK-exit rate and exit-stage NOK-recall (the fraction of defective images that survive the exit); the integrated cascade reports end-to-end NOK-recall, mean/p95/p99 per-image latency under a single strict-serial protocol (§4), and the routed-area ratio — the per-image union of inspected regions divided by image area, averaged with exited images counted as zero — which makes selectivity auditable. Two metric regimes are kept strictly separate: headline metrics are threshold-free, and every deployed threshold (exit threshold, decision-layer thresholds) is calibrated on validation only and applied frozen to test, with realised test rates published unclamped.

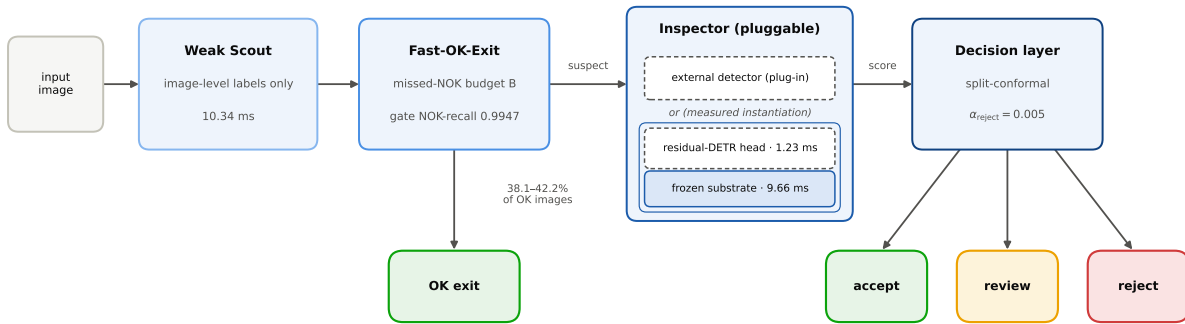
Notation. The symbols used throughout:

Symbol	Meaning
$\hat{a}(I)$	per-image anomaly score emitted by any model in the common interface
τ_{exit}	exit threshold, calibrated on validation under the missed-NOK budget
B	operator-chosen missed-NOK budget for exit calibration (evaluated at 0% and $\leq 1\%$)
c_s, c_i	per-image cost of the scout and of the routed inspection stage
e	OK-exit rate (fraction of images admitted OK at the exit)
$\alpha_{\text{accept}}, \alpha_{\text{reject}}$	conformal strictness levels of the decision layer
n_{cal}	number of validation OK scores used for conformal calibration

Table 1 — Notation. Every quantity is either measured (§5–§7) or a calibration input fixed on validation.

3. Method: a risk-calibrated selective-inspection cascade

Figure 1 shows the cascade. One production image traverses it as follows: the weak scout scores the image; if the score falls below the calibrated exit threshold, the image is admitted OK on the spot and no further compute is charged; otherwise the image is forwarded to the inspector, whose score the subordinated conformal layer turns into the line’s accept, review, or reject decision. All deployment knobs — the exit budget B , the decision-layer strictness α , and the inspector choice — are post-hoc: like the flexible speed tuning of real-time detection transformers (Zhao et al. 2024), they reconfigure a deployed line without retraining.



Stage timings from a single strict-serial latency protocol (RTX 4090, 256 px, batch 1); rates measured on the leakage-clean $n = 802$ test surface.

Figure 1: The risk-calibrated selective-inspection cascade: a weak scout trained from image-level labels only, a Fast-OK-Exit stage calibrated under a missed-NOK budget, and a pluggable inspector (an external detector plugs in unchanged; the measured integrated instantiation is a residual-DETR head over a frozen distillation substrate), completed by the subordinated split-conformal accept/review/reject decision layer. Stage timings come from a single strict-serial latency protocol (§4); rates from the leakage-clean $n=802$ test surface.

3.1 The weak scout

The scout (`scout_image_bce`) is a DETR-lite (Carion et al. 2020) weak head over an ImageNet-pretrained ResNet-18 (He et al. 2016) at 256×256 input, whose global-pooled image score is trained with binary cross-entropy on the image-level OK/NOK label. Image labels are the only supervision consumed — no boxes, no masks — so the scout is trainable from the labels a line accumulates first, and the same trained head supplies the exit scores of §3.2. How far image labels alone carry it, and where they stop, is the subject of §5.

3.2 Risk-calibrated Fast-OK-Exit

Most inline images are normal, so the cheapest useful decision is to admit confident OK images early and spend compute only where suspicion remains — the economics that make millisecond-class inspection an explicit design goal in OK-only anomaly detection (Batzner, Heckler, and König 2024), and that coarse-to-fine cascades exploit by avoiding computation on unremarkable content (Yang, Huang, and Wang 2022; Zheng et al. 2023). The exit applies a **recall-guarantee rule**: the exit threshold τ_{exit} is the largest threshold that keeps wrongly-exited validation NOK images within the operator-chosen missed-NOK budget B (we evaluate $B = 0\%$ and $B \leq 1\%$), calibrated on validation only and applied frozen to test. Images scoring below τ_{exit} are admitted OK and bypass all downstream compute. The calibration is deliberately simple — a one-dimensional threshold under an interpretable constraint — because the operator-facing contract matters more than the mechanism: B is the single number a line manager is asked to choose.

3.3 The pluggable inspector

The inspector is deliberately external to the contribution: any model that emits a per-image score (and optionally region proposals) in the shared prediction interface can serve, and the cascade claims none of the inspector’s accuracy as its own. The strongest plug-in we measured is the fully supervised RF-DETR-S (Robinson et al. 2026) at 0.9977 image-AUROC on our surface; a distilled student of it (`t2b_distilled`, 0.9614) offers an own-weights alternative whose latency and parameter count we have not measured and for which we therefore make no smaller-or-faster claim. The **measured integrated instantiation** in §6 uses our residual-DETR head: a tiny DETR decoder (16 queries, 2 layers) over residual-peak tokens from a frozen OK-only distillation substrate (Deng and Li 2022), trained head-only. We state its standing honestly where it matters: the head’s own image-level score (0.8216, 95% CI [0.7862, 0.8539]) never exceeds its frozen substrate’s (map-max 0.8424; substrate standalone 0.8465), so it contributes region proposals and per-region score traces rather than image-level discrimination, and it — not the exit stage — is the integrated cascade’s recall bottleneck (§6.2). The exit-stage metrics of §6.1 are independent of this choice; a stronger plug-in changes the integrated recall, not the exit behaviour.

3.4 The subordinated conformal decision layer

Downstream of the cascade, a split-conformal (Vovk, Gammerman, and Shafer 2005; Angelopoulos and Bates 2023) accept/review/reject layer converts the operative scorer’s image score into the line’s three-way decision. On the $n_{\text{cal}} = 614$ validation OK scores, $\tau(\alpha)$ is the k -th smallest with $k = \lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ — accept below $\tau(\alpha_{\text{accept}})$, reject above $\tau(\alpha_{\text{reject}})$, review in between. Under exchangeability the reject rate carries the marginal finite-sample guarantee $P(\text{OK rejected}) \leq \alpha_{\text{reject}}$, certifiable only for $\alpha \geq 1/(n_{\text{cal}} + 1) = 0.001626$; we instantiate $\alpha_{\text{accept}} = 0.1$ and $\alpha_{\text{reject}} = 0.005$, applied frozen to test. The layer is subordinated — it is an operating layer over whatever scorer the cascade ends in, not a detector — and scorer-agnostic: the same rule, with the same guarantee, calibrates over own and external scorers alike (§7), making acceptance strictness a post-hoc, risk-controlled knob (Angelopoulos et al. 2024).

4. Experimental protocol

4.1 Dataset and surfaces

All training and scoring is on MVTec AD (Bergmann et al. 2019) (15 texture and object categories), re-carved from its original unsupervised form into the label pools of §2 — so our numbers are comparable to the original unsupervised leaderboard only for the label-free floor rows. Every model is scored on one shared held-out test surface of $n = 802$ images (614 OK / 188 NOK), built with seed 1337, so every comparison in this paper shares one axis; a companion validation surface of the same construction and composition is the sole calibration surface for every threshold, and the test surface is never used for any form of tuning. Each label pool trains from its own manifest, and all pools are disjoint from the shared test surface.

4.2 Disjointness, machine-checked

Disjointness is enforced at the **image-identifier level** — renamed or re-exported copies of the same image count as overlap, so file-name checks are not sufficient. The rule is absolute: an evaluation surface is contaminated for training the moment it is used to score any model, and may never afterwards appear, in whole or in part, in any model’s training split. Every reported cell carries a machine-generated acceptance record that re-runs the image-identifier intersection checks (train–evaluation, train–prediction, prediction–evaluation) and stores their outcomes alongside the training manifests, the evaluation manifest, and pointers to the registered run behind the cell; a cell without a passing record cannot support a claim. The $\text{train} \cap \text{test}$ intersection of every pool against the $n = 802$ surface is zero. A small set of earlier cross-dataset measurements, taken under the same per-dataset disjointness checks before the evaluation was consolidated onto a single benchmark, is reported once, as clearly labelled secondary evidence (§5.3).

4.3 Baselines and training

Weak-label baselines are YOLO11 classification models (Jocher and Qiu 2024) trained on the identical pools, so every row of Table 2 consumes the same annotation budget as the scout. The label-free floor is PatchCore (Roth et al. 2022) and companions trained on a pooled 15-category OK set (one model for all categories) under reference recipes (Akay et al. 2022); pooled-protocol numbers sit below the per-category numbers those methods publish, and the comparison remains internally fair because every method faces the same pooled protocol. The scout trains single-seed at seed 1337 except in §5.2, where seeds 2027 and 4242 replicate the supervision-budget curve. The residual-DETR inspector head of §3.3 trains head-only: its substrate checkpoint is frozen and bit-identical throughout, so the head’s contribution is separable from the substrate’s by construction.

4.4 Latency protocol (strict-serial)

All experiments run on a single NVIDIA RTX 4090 in fp32 (PyTorch eager); scoring and latency measurement run entirely on the GPU. Every latency number in this paper comes from one protocol: batch 1, 256 px input, 50-image warmup, strict-serial execution with zero GPU co-tenancy verified at run time, CUDA-synchronize bracketing every timed region (per-stage preprocessing and host–device transfers inside the brackets, disk I/O outside), five timed sweeps with per-image medians, and selective-path timing replaying the frozen stored scout scores that define the exit decisions (zero decision flips between replay and live execution). We never place a figure from this protocol next to a timing measured under any other protocol, batch size, resolution, or device — published timings of external methods are not latency-comparable to ours and do not appear alongside them.

4.5 Statistics

Confidence intervals are 95% bootstrap intervals over the test surface, except binomial proportions (exit-stage NOK-recall), which carry exact Clopper–Pearson 95% intervals; both capture test-resampling variance only. Training variance is addressed where it matters most — the supervision-budget curve carries a three-seed replication (§5.2) — and single-seed elsewhere, stated as such. No paired tests or multiple-comparison

corrections are applied, and cross-method AUROC margins smaller than about 0.01 are read as indistinguishable, not as rankings — a rule we apply to our own results first (§5.1). All metrics are recomputed from stored prediction files, and the integrated cascade of §6 is composed from those stored predictions by a checkable generator, numerically identical to a live execution.

5. How many defect labels are enough?

5.1 The scout against its weak-label competitors and the label-free floor

Table 2 compares the scout against image-label classifiers trained on the identical pools, and against the label-free floor that consumes no NOK labels at all. The scout leads both weak-label pools among methods consuming image labels: 0.9339 vs 0.8571 (YOLO11-S) and 0.7250 (YOLO11-N) on the few-NOK pool; 0.9110 vs 0.8532 and 0.8252 on the balanced pool. The floor comparison is where honesty is owed, and we state it plainly. On the few-NOK pool the scout’s margin over PatchCore (0.9242) is smaller than 0.01 — indistinguishable under our own reading rule (§4); on the balanced pool the floor is *not cleared* (0.9110 < 0.9242). The scout’s case is therefore not ranking accuracy against label-free methods: it is that image labels buy a *calibratable* exit score (§6) at an annotation cost — image-level labels over the same pools where localisation-based training would require boxes or masks (Božič, Tabernik, and Skočaj 2021) — that the line has already paid.

Method (supervision consumed)	few-NOK pool	balanced pool
Weak scout (ours; image labels only)	0.9339	0.9110
YOLO11-S classifier (image labels)	0.8571	0.8532
YOLO11-N classifier (image labels)	0.7250	0.8252
PatchCore (label-free floor; no NOK labels)	0.9242	0.9242

Table 2 — Image-AUROC on the shared $n=802$ test surface. The label-free floor row trains on OK images only and ignores both pools’ NOK labels; its number is identical in both columns by construction. Bold marks the only margin exceeding 0.01 against all rows of its column.

5.2 The supervision-budget curve

Figure 2 and Table 3 answer the deployment question directly: with the OK pool fixed at 2 361 images and only the NOK count varied, image-AUROC moves $25 \rightarrow 0.583$, $50 \rightarrow 0.512$ (bootstrap CI [0.464, 0.560], straddling chance), $100 \rightarrow 0.749$, $200 \rightarrow 0.860$, $400 \rightarrow 0.911$, and 0.9339 at the full 727-NOK budget. Three findings, each checked across three training seeds: the near-chance floor at ≤ 50 NOK **replicates**; the rescue at ≥ 200 NOK **replicates** (cross-seed range ≤ 0.042); and the 100-NOK knee is **seed-dependent** — outcomes span 0.569–0.810 (range 0.241). A deployment plan should therefore budget for at least 200 labelled defect images, not roughly 100, and the replicated claim is the curve’s *shape*, not any individual transition point.

The mechanism behind the floor is informative rather than surprising: below 50 NOK examples the positive class is too thin for the image-level loss to shape a usable decision boundary, and the balanced-pool comparison of §5.3 shows the same lever from the other side — when NOK examples are scarce, re-balancing what exists is worth more than any architectural change we measured. The curve is the piece a deployment plan can act on before any hardware is bought: it converts “collect more defect labels” from advice into a budget with a measured payoff schedule.

NOK labels in the pool	image-AUROC (primary seed)	across three training seeds
25	0.583	near-chance floor — replicates
50	0.512 (CI [0.464, 0.560])	near-chance floor — replicates
100	0.749	seed-dependent: 0.569–0.810 (range 0.241)
200	0.860	rescue — replicates (cross-seed range ≤ 0.042)
400	0.911	rescue — replicates
727 (full pool)	0.9339	rescue — replicates

Table 3 — The supervision-budget curve, point by point: image-AUROC on the fixed $n=802$ test surface with the OK pool held at 2 361 images and only the NOK count varied. The right column states what the three-seed replication supports at each point; per-point values inside the seed-dependent transition region should not be relied on.

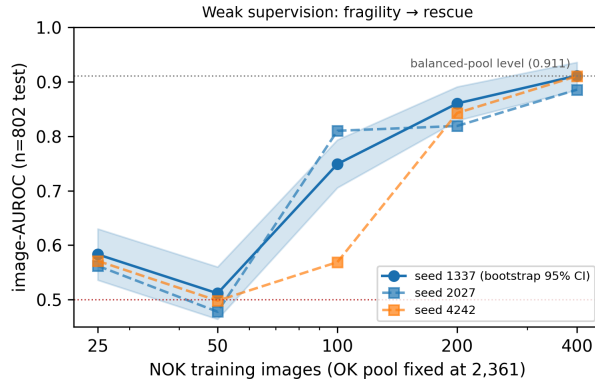


Figure 2: Supervision-budget curve for the weak scout: image-AUROC on the fixed $n=802$ test surface vs NOK training count, with the OK pool fixed at 2 361. Solid: primary seed with bootstrap 95% CI band; dashed: replication seeds. The ≤ 50 floor and the ≥ 200 rescue replicate; the 100-NOK knee is seed-dependent (0.569–0.810). Reference lines: chance (0.5) and the balanced-pool level (0.9110).

5.3 Secondary evidence: pool balance across benchmarks

Before the evaluation was consolidated onto the single benchmark of §4, the same scout recipe was measured on four further public benchmarks under the same per-dataset image-identifier disjointness checks; we report those cells once, here, as secondary evidence, each on its own per-dataset test surface rather than the $n=802$ surface. With balanced pools the scout reached 0.9659 on MPDD (Jezek et al. 2021), 0.8760 on MVTec AD 2 (Heckler-Kram et al. 2025), 0.8289 on VisA (Zou et al. 2022), and 0.8700 on Real-IAD (Wang et al. 2024); with few-NOK pools the same recipe collapsed to 0.5451 (VisA) and 0.5933 (Real-IAD). Two readings follow. Pool balance, not architecture, was the single biggest driver of weak-label viability across benchmarks — balanced sampling rescued every surface. And the few-NOK operating point that succeeds on the primary benchmark does not transfer for free: on harder surfaces the scout needs either more NOK labels or a rebalanced pool, which bounds the scope of §5.2’s curve to the benchmark it was measured on.

6. The measured risk–compute trade-off

6.1 Exit calibration: what the scout’s score buys

Calibrated on validation and applied frozen to test, the $B = 0\%$ missed-NOK budget exits 234/614 OK images (38.1%) and the $B \leq 1\%$ budget exits 259/614 (42.2%), both at an exit-stage NOK-recall of 0.9947 — 187 of 188 defective images survive the exit (Clopper–Pearson 95% CI [0.9707, 0.9999]). These numbers are properties of the scout and its calibration alone: they hold for any inspector plugged in downstream.

Relaxing the budget from 0% to $\leq 1\%$ buys 4.1 points of OK-exit at no measured cost in test exit-stage recall — though at single-miss resolution this shows only that the two budgets are indistinguishable on this surface, not that the band is provably flat.

6.2 The integrated cascade, measured as one system

The end-to-end numbers require an inspector, and we state the instantiation honestly: the integrated measurement uses our own residual-DETR head over its frozen distillation substrate (§3.3), not the strongest available plug-in. Under that instantiation, with the inspector’s validation-calibrated threshold applied at full precision, the integrated cascade detects 0.8670 of NOK images at both exit budgets, against 0.8723 for the full path that runs the inspector on everything: the integration costs exactly **one** exit-stage-missed defect the inspector would have caught, and the remaining 24 misses belong to the inspector itself (miss attribution: 1 exit stage + 24 inspector). The routed-area ratio is 0.0098 under the region-union definition of §2 — the cascade inspects about 1% of pixel area on average, exited images counted as zero. The inspector, not the exit policy, is the recall bottleneck; a stronger plug-in attacks the 24, not the 1.

One implementation detail earned its place in the results because it silently changed them: thresholds must be applied *exactly* as calibrated. Rounding the inspector’s calibrated threshold at the fourth decimal dropped a further borderline defect (0.8617 instead of the correct 0.8670) with no error raised anywhere — in a regime where single images move the headline, threshold hygiene is a measurement issue, not a style preference.

6.3 Cost accounting, and the honest reading

Table 4 gives the per-image cost under the single strict-serial protocol of §4. The full path costs 21.22 ms on average (scout 10.34 + substrate 9.66 + head 1.23); the selective path costs 18.03 ms at the 0% budget and 17.69 ms at $\leq 1\%$ — a 15.1% / 16.7% mean-cost reduction.

Path (single strict-serial protocol, §4)	mean (ms)	p95 (ms)	Δ mean
Full: scout 10.34 + substrate 9.66 + head 1.23	21.22	22.20	—
Selective, $B=0\%$ / $B\leq 1\%$ exit budget	18.03 / 17.69	22.15 / 22.13	−15.1% / −16.7%

Table 4 — Intra-protocol per-image cost (RTX 4090, 256 px, fp32, batch 1, strict-serial). No cross-protocol timing appears in this paper.

The honest reading: **the exit buys throughput, not detection latency**. The scout runs on every image and an exit saves only the routed-stage cost, so the saving lands on the mean while p95/p99 are essentially unchanged (≈ 22.2 ms) — the tail is routed by construction, and every detected NOK rides the routed path. The recall-at-latency view (Figure 3) makes the same point from the detection side: the selective path’s normalized AUC is 0.0931 against 0.3538 for the full path, because an exited NOK contributes zero recall at any latency budget and every detection still pays the full scout + substrate + head chain. The measured effect of the exit is therefore a 15.1–16.7% mean per-image cost saving at a recall cost of exactly one NOK (0.8670 vs 0.8723) — not a latency-class change, and we make no claim that it is one.

7. The subordinated decision layer

The conformal layer of §3.4 is evaluated exactly as defined: accept/review/reject thresholds calibrated on the 614 validation OK scores, applied frozen to test, realised rates published unclamped, at $\alpha_{\text{reject}} = 0.005$ — above the certifiable floor of $1/(n_{\text{cal}} + 1) = 0.001626$. Scorer-agnosticism is the claim: Table 5 applies the

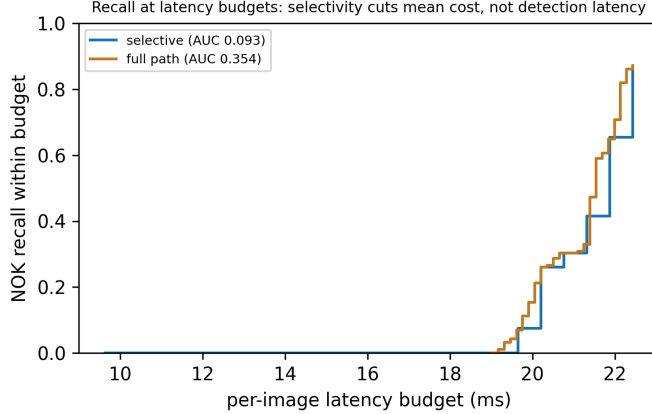


Figure 3: Recall within a per-image latency budget at the validation-calibrated operating threshold (all timings from the strict-serial protocol: RTX 4090, 256 px, batch 1, fp32). The selective path (AUC 0.0931) holds zero recall until ≈ 19.7 ms because an exited NOK contributes zero recall and every detection still pays the full chain; the full path (AUC 0.3538) accumulates recall earlier. The exit buys mean per-image cost (throughput), not detection latency.

same rule, carrying the same finite-sample marginal false-reject guarantee, over four scorers — the strongest external plug-in, our distilled student, the few-NOK scout itself, and the label-free floor — on the common surface. The review band buys the false-alarm reduction a single threshold cannot, at an explicit inspection cost, and the layer’s strictness is retunable after deployment without touching any scorer.

Scorer under the identical rule ($\alpha_{\text{reject}}=0.005$)	FA/10k OK	missed-defect rate	review rate
RF-DETR-S (external plug-in)	32.6	0.0053	8.2%
Distilled student (ours)	65.1	0.0798	13.3%
Weak scout, few-NOK (ours)	65.1	0.2074	13.5%
PatchCore (label-free floor)	32.6	0.2713	11.3%

Table 5 — Certifiable conformal accept/review/reject operating points, applied identically over own and external scorers on the common $n=802$ surface; realised test rates published unclamped.

A separate validation-calibrated read-out sharpens the same ranking at low false-alarm budgets: the external plug-in keeps 0.968 recall at the $\text{FA} \leq 100/10\text{k}$ budget (realised 97.7/10k) and 0.9255 at $\text{FA} \leq 1$; the distilled student retains 0.761; the few-NOK scout and the label-free floor sit near 0.59. The decision layer does not create discrimination — it converts whatever discrimination the operative scorer has into a guaranteed operating contract, which is precisely its subordinated role in the cascade.

8. Deployment guidance: reading the measurements as an operating manual

The measurements of §5–§7 were designed to answer the questions a deployment actually asks, so we close the results with the answers in that form. Each recommendation below cites the measurement that grounds

it; none extrapolates beyond the evaluated surface.

How many defect labels should be collected before trusting a weak scout? Budget for at least 200 image-level NOK labels. Below 50 the scout is statistically indistinguishable from chance (0.512 at 50, CI straddling 0.5); at 100 the outcome is a seed lottery (0.569–0.810 across three seeds); at 200 it is reliably usable (0.860, replicated within a 0.042 cross-seed range) and improves smoothly thereafter (§5.2). If the line can curate the retraining pool, balance it: on every surface where both pools were measured, balance was worth more than architecture (§5.3).

How should the exit budget B be chosen? Start at $B = 0\%$: it already exits 38.1% of OK images at 0.9947 exit-stage NOK-recall. Relaxing to $B \leq 1\%$ bought 4.1 points of additional exit at no measured test recall cost on our surface (§6.1) — but that equivalence was established at single-miss resolution, so treat the relaxation as a throughput knob to be re-validated on the line’s own validation stream, not as a free lunch. The calibration is a one-dimensional threshold on a held-out OK/NOK score sample; recalibrating after drift costs one validation pass, no retraining.

What should the exit be expected to buy? Mean per-image cost and throughput — nothing else. On our instantiation the saving was 15.1–16.7% of mean cost with p95/p99 unchanged (§6.3). Provisioning, takt-time, and worst-case-latency decisions must be made against the *full-path* latency, because every detected defect rides the full path; the exit only thins the normal traffic around it. The relative saving also depends on the cost ratio: with a scout at roughly half the total path cost (10.34 of 21.22 ms), the exit can at best halve the mean — a cheaper scout or a costlier inspector widens the achievable saving, a relation the cost model of §2 makes explicit.

Which inspector should be plugged in? The strongest available one: the cascade claims none of the inspector’s accuracy, and the integrated recall is inspector-limited (24 of 25 misses were inspector misses under our own-head instantiation; §6.2). Where an external detector is deployable, its measured plug-in quality (0.9977 image-AUROC supervised) makes it the default choice; the exit-stage guarantees are unaffected by the swap.

How should the accept/review/reject thresholds be run? Through the conformal layer, not hand-set thresholds: it converts the operative scorer’s scores into a three-way decision whose false-reject rate carries a finite-sample guarantee, certifiable down to $\alpha = 0.001626$ with our 614-image calibration set (§7). Strictness changes are post-hoc; a change of scorer requires only recalibration on the same validation OK scores. Size the calibration set before promising a budget: the smallest certifiable α is $1/(n_{\text{cal}} + 1)$, so a line that must certify a 0.1% false-reject budget needs a thousand-image OK calibration set, not a better model.

9. Related work

Selective and coarse-to-fine inference. QueryDet (Yang, Huang, and Wang 2022) accelerates detection by avoiding computation on background regions, and Focus-DETR (Zheng et al. 2023) makes the computation–accuracy trade-off the contribution itself; our cascade descends from this lineage but moves the selection decision to the *image* level, trains it from weak labels only, and — the part this literature leaves open — calibrates it under an explicit missed-defect budget with the realised risk measured end-to-end.

OK-only industrial anomaly detection. The dominant industrial paradigm trains on normal images alone — PatchCore (Roth et al. 2022), Reverse Distillation (Deng and Li 2022), EfficientAD (Batzner, Heckler, and König 2024), whose efficiency-as-economics framing our exit stage follows. On our surface this regime sets the label-free floor the scout must justify itself against, and we concede the comparison where it is not won (§5.1): the scout’s value is the calibratable exit score, not the ranking.

Weak and mixed supervision. The KolektorSDD2 study (Božič, Tabernik, and Skočaj 2021) traces the weakly-to-fully-supervised path in surface-defect detection that our supervision-budget curve operationalises with a controlled NOK-count sweep and seed replication; to our knowledge the curve’s calibrated-exit consequence — how many image-level defect labels a *safe early-exit policy* needs — had not been measured.

Real-time detection transformers. The real-time DETR line — RT-DETR (Zhao et al. 2024), RT-

DETRv2 (Lv et al. 2024), RF-DETR (Robinson et al. 2026) — supplies deployment-grade supervised detectors with post-hoc speed tuning. For this paper that line supplies *inspectors*: the cascade absorbs the strongest one we measured as a plug-in and claims none of its accuracy.

Conformal prediction. Split-conformal calibration provides finite-sample, distribution-free guarantees (Vovk, Gammerman, and Shafer 2005; Angelopoulos and Bates 2023), extended to bounded risks by conformal risk control (Angelopoulos et al. 2024). Our decision layer applies the split-conformal rule to image scores from any scorer; its contribution here is the subordinated-layer packaging — one certifiable knob over an interchangeable scorer — not new conformal theory.

Evaluation practice. Recent critical analyses argue that academic protocols misstate production behaviour in industrial anomaly detection (Baitieva et al. 2025; Cheng et al. 2025); our protocol (§4) — machine-checked image-identifier disjointness, frozen calibrated thresholds, strict-serial latency, unclamped realised rates — is a single-system application of that critique to ourselves, on the benchmark of (Bergmann et al. 2019).

10. Limitations

The scout’s comparison set is incomplete. The weak-label baselines measured are the YOLO11 classifiers of Table 2. Two natural head-to-heads remain unmeasured: a plain ResNet-18 classifier (the scout minus its DETR-lite neck), which would isolate whether the neck contributes anything over the backbone, and a classic classifier→detector cascade built from an off-the-shelf classifier, which would price our calibration rule against the obvious engineering alternative. Nothing in this paper establishes the DETR-lite neck as necessary; both comparisons are future work, and until they run, the defensible claim is the calibrated-exit result, not scout-architecture superiority.

Single primary benchmark, with bounded secondary evidence. Every headline number is measured on one benchmark and one held-out surface (§4). The cross-benchmark cells of §5.3 are secondary evidence on per-dataset surfaces, and they cut both ways: they replicate the balance finding, and they show the few-NOK operating point failing to transfer to two of four other surfaces. No result here is claimed beyond the surface it was measured on.

The supervision-budget curve’s transition point is seed-dependent. The floor (≤ 50 NOK) and the rescue (≥ 200 NOK) replicate across three seeds (cross-seed range ≤ 0.042), but the 100-NOK point spans 0.569–0.810 (range 0.241): per-point values in the transition region should not be relied on, and the claim is the curve’s shape (§5.2). On the balanced pool the label-free floor is not cleared at all ($0.9110 < 0.9242$), and the few-NOK margin over that floor is within our indistinguishability rule.

The selective benefit is throughput, not tail or detection latency — and it is protocol-bound. The measured saving is a mean-cost effect only: -15.1% / -16.7% at the two exit budgets, with p95/p99 essentially unchanged and the selective path accumulating recall later than the full path (AUC 0.0931 vs 0.3538; §6.3). All latency was measured under one strict-serial protocol on one device; transfer to other hardware, batch sizes, or resolutions is not characterised.

The integrated instantiation is inspector-limited. The end-to-end recall (0.8670) reflects the residual-DETR head instantiation, whose image-level score never exceeds its own frozen substrate ($0.8216 \leq 0.8424 \leq 0.8465$) and which accounts for 24 of the 25 integrated misses. The exit-stage results (§6.1) are inspector-independent, but the end-to-end numbers should be read as a lower bound an off-the-shelf stronger plug-in would lift; measuring the integrated cascade with the external inspector, and profiling the distilled student’s latency and parameters, are future work. The cascade makes no localisation-quality claim for any component: region proposals serve routing and evidence, not measurement.

The conformal guarantee is marginal and exchangeability-bound. $P(\text{OK rejected}) \leq \alpha_{\text{reject}}$ holds on average over calibration draws — not per category or subgroup — and distribution shift voids it. With 614 calibration OK images the smallest certifiable level is 0.001626; meaningfully tighter budgets require more validation OK data, not a different estimator.

11. Conclusion

The question was falsifiable and the answer is measured: a scout trained only with image-level OK/NOK labels can safely discard a large fraction of normal images before an expensive inspector runs, under a calibrated missed-defect budget — 38.1–42.2% of OK images exit at 0.9947 exit-stage NOK-recall, for a 15.1–16.7% mean per-image cost reduction whose honest name is throughput, not latency, at a measured integration cost of exactly one defect. The supervision-budget curve turns the result into deployment guidance: budget for at least 200 labelled defects, distrust the seed-dependent region near 100, and expect nothing below 50. The subordinated conformal layer converts whichever scorer ends the cascade into a certifiable accept/review/reject contract, and every deployment knob — exit budget, strictness, inspector choice — moves without retraining.

What the paper does not claim is as deliberate as what it does: no accuracy improvement over any plugged-in detector, no clearing of the label-free floor on the balanced pool, no tail-latency gain, and no localisation quality. The next measurements are named in §10 — the plain-classifier head-to-head, the classic-cascade baseline, and the integrated cascade under its strongest plug-in. A production line deploys an operating point, not a leaderboard entry; we publish ours, with its costs attached.

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